

NexaCloud

API Reference Guide

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This document provides a complete reference for the NexaCloud REST API. It covers authentication, all available endpoints, request/response formats, error handling, and code examples in Python, JavaScript, and cURL. The NexaCloud API is RESTful, uses JSON for request/response bodies, and authenticates via Bearer tokens. All requests must be made over HTTPS.

Property	Value
Base URL	https://api.nexacloud.io/v1
Protocol	HTTPS only (TLS 1.2+)
Format	JSON (application/json)
Authentication	Bearer Token (API Key)
Rate Limiting	Per plan (see Rate Limiting Guide API-002)
Versioning	URL path versioning (/v1, /v2)
Pagination	Cursor-based (page + per_page)
Timestamps	ISO 8601 / UTC (YYYY-MM-DDTHH:MM:SSZ)

1. Authentication

1.1 API Key Authentication

All NexaCloud API requests require authentication using an API key passed as a Bearer token in the Authorization header. API keys are scoped to specific permissions and can be created, managed, and revoked from the Settings → API Keys dashboard.

Request Header Format:

```
Authorization: Bearer YOUR_API_KEY Content-Type: application/json Accept: application/json
```

Python Example:

```
import requests API_KEY = "nxc_live_your_api_key_here" BASE_URL = "https://api.nexacloud.io/v1" headers = { "Authorization": f"Bearer {API_KEY}", "Content-Type": "application/json" } response = requests.get(f"{BASE_URL}/projects", headers=headers) print(response.json())
```

cURL Example:

```
curl -X GET "https://api.nexacloud.io/v1/projects" \ -H "Authorization: Bearer YOUR_API_KEY" \ -H "Content-Type: application/json"
```

1.2 API Key Scopes

Scope	Access Level	Recommended Use Case
read:all	Read all resources	Dashboards, reporting tools
write:projects	Create/update projects	CI/CD integrations
write:users	Manage users	User provisioning systems
write:data	Write data sources	ETL pipelines
admin	Full access	Backend admin systems (use with caution)
billing:read	View billing info	Finance integrations
webhooks	Manage webhooks	Event-driven integrations

■ ■ Security Note: Always use the minimum required scope. Rotate API keys every 90 days. Never commit API keys to source control — use environment variables or secrets management tools.

2. Projects API

Method	Endpoint	Description	Scope Required
GET	/v1/projects	List all projects	read:all
POST	/v1/projects	Create a new project	write:projects
GET	/v1/projects/{id}	Get project by ID	read:all
PUT	/v1/projects/{id}	Update project	write:projects
DELETE	/v1/projects/{id}	Delete project	admin
GET	/v1/projects/{id}/members	List project members	read:all
POST	/v1/projects/{id}/members	Add member to project	write:projects

2.1 List Projects

GET /v1/projects

Returns a paginated list of all projects in your organization.

Query Parameters:

Parameter	Type	Required	Description
page	integer	No	Page number (default: 1)
per_page	integer	No	Items per page (default: 20, max: 100)
status	string	No	Filter: active, archived, all (default: active)
search	string	No	Search by project name
sort	string	No	Sort field: name, created_at, updated_at
order	string	No	Sort order: asc, desc (default: desc)

Response (200 OK):

```
{ "data": [ { "id": "proj_abc123", "name": "Marketing Analytics Q2", "description": "Q2 campaign performance tracking", "visibility": "private", "status": "active", "owner": { "id": "usr_def456", "name": "Jane Smith", "email": "jane@company.com" }, "member_count": 5, "created_at": "2025-01-15T10:30:00Z", "updated_at": "2025-06-01T08:15:00Z" } ], "pagination": { "total": 42, "page": 1, "per_page": 20, "total_pages": 3, "has_next": true, "next_cursor": "cursor_xyz789" } }
```

3. Users API

Method	Endpoint	Description	Scope Required
GET	/v1/users	List organization users	read:all
POST	/v1/users/invite	Invite a new user	write:users
GET	/v1/users/{id}	Get user details	read:all
PUT	/v1/users/{id}	Update user profile/role	write:users
DELETE	/v1/users/{id}	Remove user from org	write:users
POST	/v1/users/{id}/deactivate	Deactivate user	write:users
GET	/v1/users/me	Get current user	read:all

3.1 Invite User

POST /v1/users/invite

Send an invitation email to a new team member.

Request Body:

```
{ "email": "newmember@company.com", "role": "member", "projects": ["proj_abc123", "proj_def456"], "send_welcome_email": true }
```

Response (201 Created):

```
{ "data": { "id": "inv_ghi789", "email": "newmember@company.com", "role": "member", "status": "pending", "invited_by": "usr_def456", "expires_at": "2025-06-08T10:00:00Z" } }
```

4. Data Sources API

Data Sources are connections to external databases, APIs, and services. Once connected, data is automatically synced based on the configured schedule.

Method	Endpoint	Description
GET	/v1/data-sources	List all data sources
POST	/v1/data-sources	Create a new data source
GET	/v1/data-sources/{id}	Get data source details
PUT	/v1/data-sources/{id}	Update data source config
DELETE	/v1/data-sources/{id}	Remove data source
POST	/v1/data-sources/{id}/sync	Trigger manual sync
GET	/v1/data-sources/{id}/sync-logs	View sync history
POST	/v1/data-sources/test	Test connection before saving

4.1 Create PostgreSQL Data Source

POST /v1/data-sources

```
{ "name": "Production Database", "type": "postgresql", "config": { "host": "db.company.com", "port": 5432, "database": "prod_analytics", "username": "nexacloud_reader", "password": "encrypted_password", "ssl_mode": "require" }, "sync_frequency": "hourly", "sync_tables": ["orders", "customers", "products"] }
```

4.2 Supported Data Source Types

Type Value	Integration	Authentication Method
postgresql	PostgreSQL	Username/Password + SSL
mysql	MySQL	Username/Password + SSL
bigquery	Google BigQuery	Service Account JSON
snowflake	Snowflake	Username/Password + Account
mongodb	MongoDB	Connection URI
rest_api	Custom REST API	API Key / OAuth 2.0
google_analytics	Google Analytics 4	OAuth 2.0
salesforce	Salesforce	OAuth 2.0

amazon_s3	Amazon S3	AWS Access Key + Secret
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5. Webhooks API

Webhooks enable real-time event notifications to your server. See the [Webhook Documentation \(INT-004\)](#) for the complete list of events and payload structures.

Method	Endpoint	Description
GET	/v1/webhooks	List all webhooks
POST	/v1/webhooks	Register a new webhook
GET	/v1/webhooks/{id}	Get webhook details
PUT	/v1/webhooks/{id}	Update webhook
DELETE	/v1/webhooks/{id}	Delete webhook
GET	/v1/webhooks/{id}/events	List recent events
POST	/v1/webhooks/{id}/events/{event_id}/retry	Retry failed event
POST	/v1/webhooks/{id}/test	Send test event

5.1 Create Webhook

```
POST /v1/webhooks { "url": "https://your-server.com/webhooks/nexacloud", "events":
[ "project.created", "report.generated", "alert.triggered",
"integration.sync_failed" ], "secret": "your_hmac_secret_256bit", "active": true,
"description": "Production event handler" }
```

6. Error Handling

All errors follow a consistent format. HTTP status codes indicate the class of error, while the error code provides specific details. See the Error Codes Reference (API-003) for the complete list of error codes and remediation steps.

Standard Error Response Format:

```
{ "error": { "code": "AUTH_002", "message": "Invalid API key provided.", "details":  
"The API key does not exist or has been revoked.", "request_id":  
"req_9f3a2b1c4d5e", "documentation_url":  
"https://docs.nexacloud.io/errors/AUTH_002" } }
```

HTTP Status Code Summary:

Status Code	Meaning	Common Cause
200 OK	Success	Request completed normally
201 Created	Resource created	POST request succeeded
204 No Content	Success, no body	DELETE request succeeded
400 Bad Request	Invalid request	Missing or malformed parameters
401 Unauthorized	Auth failed	Missing/invalid/expired API key
403 Forbidden	Access denied	Insufficient API key scope
404 Not Found	Resource missing	Wrong ID or resource deleted
409 Conflict	Resource conflict	Duplicate creation attempt
422 Unprocessable	Validation failed	Field validation errors
429 Too Many Req.	Rate limited	Exceeded plan rate limits
500 Server Error	NexaCloud error	Retry with exponential backoff
503 Unavailable	Maintenance	Check status.nexacloud.io

7. Official SDKs & Code Examples

7.1 Python SDK

```
pip install nexacloud-python
```

Quick Start:

```
from nexacloud import NexaCloud # Initialize client client =  
NexaCloud(api_key="YOUR_API_KEY") # List projects projects = client.projects.list()  
for project in projects: print(f"Project: {project.name} ({project.id})") # Create  
a project new_project = client.projects.create( name="My New Dashboard",  
visibility="private" ) # Invite a user client.users.invite(  
email="colleague@company.com", role="member", projects=[new_project.id] )
```

7.2 Node.js SDK

```
npm install @nexacloud/node-sdk
```

```
const { NexaCloud } = require("@nexacloud/node-sdk"); const client = new  
NexaCloud({ apiKey: process.env.NEXACLOUD_API_KEY }); // List projects const {  
data: projects } = await client.projects.list(); projects.forEach(p =>  
console.log(`${p.name}: ${p.id}`)); // Trigger sync await  
client.dataSources.sync("ds_abc123"); console.log("Sync triggered successfully");
```

Additional SDKs available: Ruby, Go, Java, PHP. Full SDK documentation at: docs.nexacloud.io/sdks

8. API Changelog

Version	Date	Changes
v2.4	June 2025	Added cursor-based pagination; new /v1/usage/rate-limits endpoint
v2.3	March 2025	Webhook retry improvements; FIDO2 key management endpoints
v2.2	Jan 2025	BigQuery and Snowflake data source support added
v2.1	Oct 2024	Custom roles API; audit log export endpoint added
v2.0	July 2024	Breaking: unified error format; new auth scopes system
v1.9	April 2024	GraphQL API beta; real-time WebSocket events (beta)
v1.8	Jan 2024	HIPAA-compliant endpoints; data residency support

Breaking changes are announced 90 days in advance via email and the status page. Subscribe to API change notifications at status.nexacloud.io.

For API support, contact api-support@nexacloud.io or visit our developer forum at community.nexacloud.io. Full interactive API documentation with live testing is available at docs.nexacloud.io/api.